Considerations for OARS Analysis using Weighted Generalized Estimating Equations

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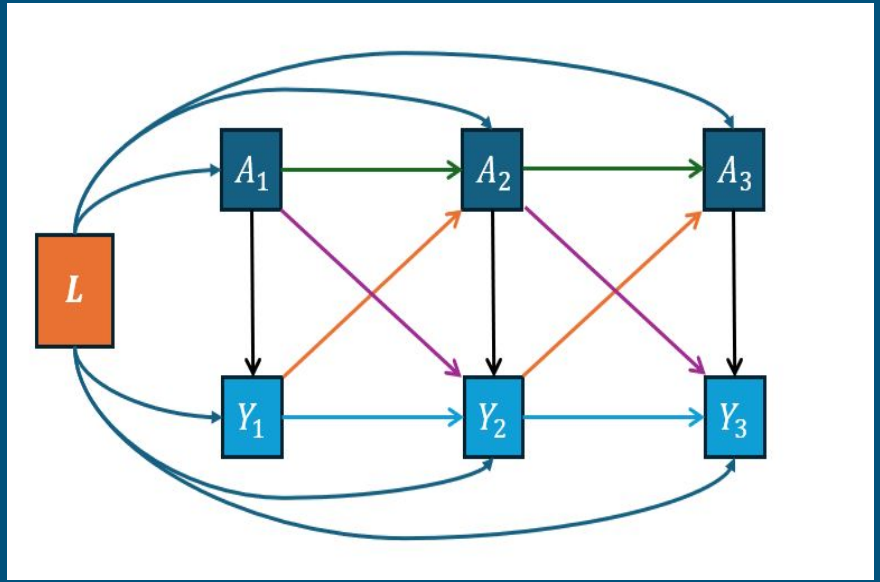
Our Agenda

1) Define the research question
2) Describe the model (Generalized Estimating Equation)
3) What is the problem of treatment switching and stopping?
4) What are the complications in modeling (e.g. rare events, longitudinal IPCW)
5) Papers and Manuscripts

Are non-opioids superior/non-inferior to opioids in managing worst pain in the OARS trial within 72 hours?

Estimate Average Treatment Effect between non-opioid and opioid group for worst pain accounting for covariates.

Based on study design (eDiary with discrete timepoints), calculate ATE at each timepoint



Generalized Estimating Equations are an appropriate model to Answer our Research Question

Family of population-level models that account for correlated (e.g. longitudinal) data and can estimate ATE at each discrete timepoint with an interaction effect.

$$E(Y_{it}|A_i, I_{it}, X_i, Z_{it}) = \beta_0 + \beta^*_{T,t=1} A_{it} + \sum_{k=2}^{6} \gamma_k I(t=k) + \sum_{k=2}^{6} \delta_k (A_i * I(t=k)) + X_i^T \beta$$

Adjusting covariates not time-dependent: correlation structure does not matter for point estimates

Treatment Variable (A) is a bit more complicated because of treatment switching and stopping

Treatment Switching and Stopping is dependent of a given patient's demographic and clinical characteristics

Day 2 am	Day 2 pm	Day 3 am	Day 3 pm	Day 4 am
95 (n=1,377)	51 (n=1,282)	85 (n=1,231)	50 (n=1,146)	10 (n=1,096)
6.8%	4.0%	6.9%	4.4%	0.9%

Censoring due to treatment switching and stopping

Combined counts of treatment switching and stopping per timepoint, as well as a percentage with those remaining

$$\boldsymbol{W} = (\text{Age, Sex, Pre-op Smoking, Clinical Site, Previous Worst Pain})$$

W has a relationship with treatment switching and stopping (censoring) and the outcome (worst pain). May need to change how we code these variables in the future

Creates an analysis problem: the data are **Missing at Random** (MAR), which GEEs can't naturally handle.

More patients in opioid group switch or stop (192 v.s. 99): likely attenuates ATE: Type II error, BAD for non-inferiority trial

Longitudinal Censoring requires techniques from marginal structural modeling

Inverse Probability of Censoring Weighting (IPCW) over time (cumulative), only on those who remain uncensored at a time k. The weights will compound over time!

$$\mathrm{SW}_{ik} = \prod_{t=1}^{k} \frac{\Pr(C_t = 0 \mid A = a_i)}{\Pr(C_t = 0 \mid A = a_i, \boldsymbol{W} = \boldsymbol{w}_i)}$$

Censoring is a Rare Event Per Timepoint

Day 2 am	Day 2 pm	Day 3 am	Day 3 pm	Day 4 am
95 (n=1,377)	51 (n=1,282)	85 (n=1,231)	50 (n=1,146)	10 (n=1,096)
6.8%	4.0%	6.9%	4.4%	0.9%

Censoring due to treatment switching and stopping

Censoring is relatively rare, and some confounders may be completely separated with censoring events: problems with ML Estimation for Logistic Regression

Adjustments: Firth's Penalization, Intercept Correction with Firth's method, logF(1,1) priors (Bayesian Methods). All have pros and cons for predictions

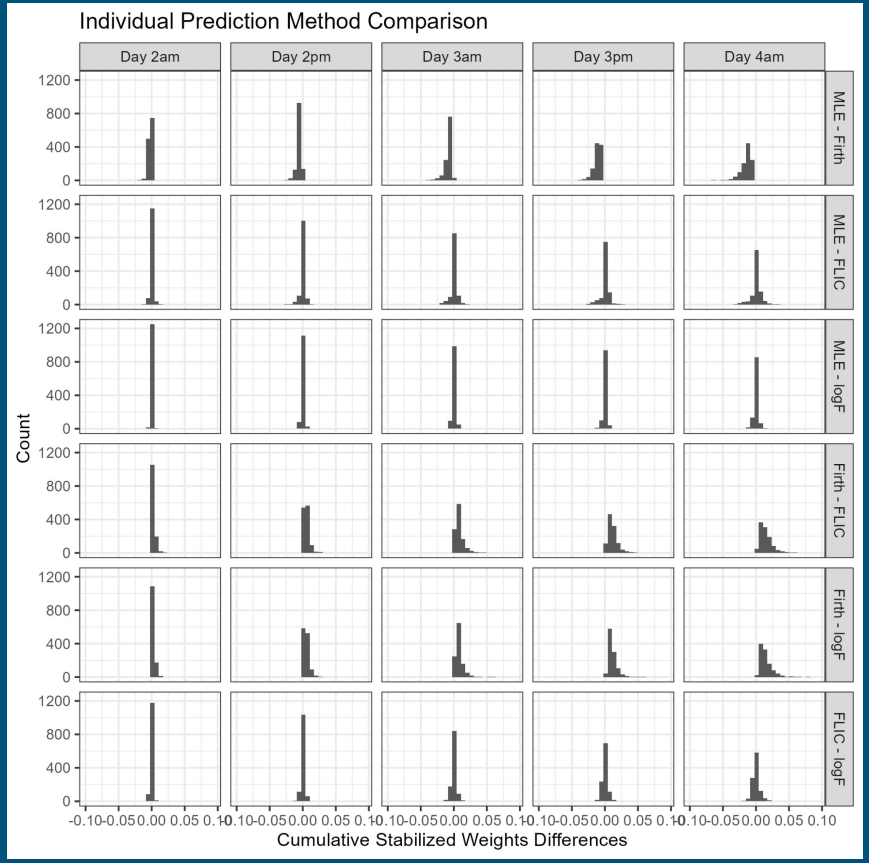
We have An Idea for Two Papers: One Methodological and One Applied

Paper 1: Simulation Study testing performance of ML Estimation, Firth, FLIC, and logF(1,1) prior for stabilized weight calculations. Compare performance with the true parameters. Not concerned about calibration, discrimination, ridge, or LASSO.

> Hypothesis: MLE, FLIC, logF(1,1) unbiased but MLE has higher variance due to separation. Firth will be somewhat biased

Paper 2: GEE Analysis of OARS for ATE, both unweighted and weighted as a sensitivity analysis and using Paper 1 results to inform which method to obtain stabilized weights.

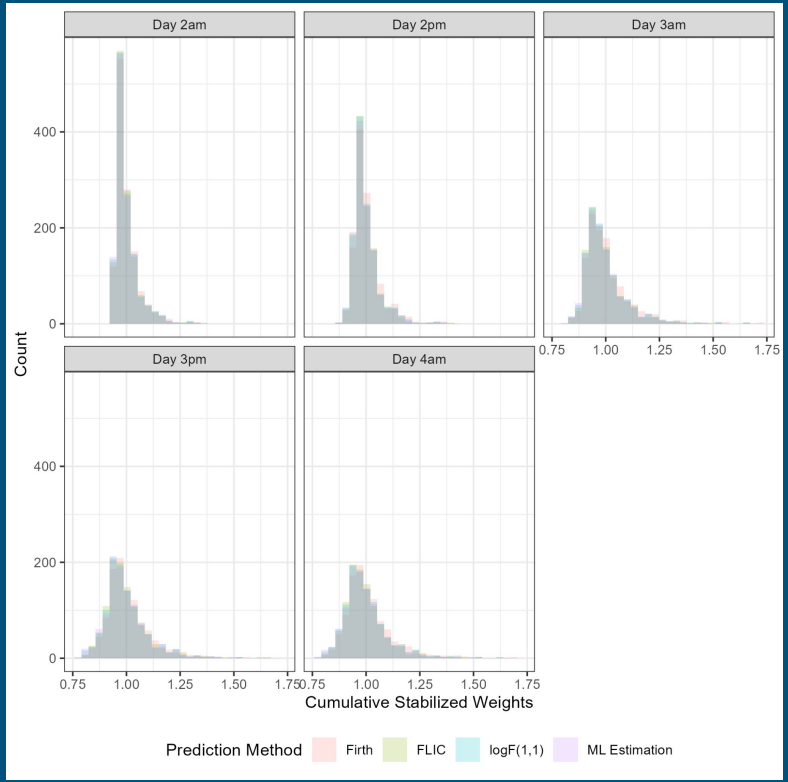
Supplementary 1



Accurate and Precise predictions for rare events logistic regression is difficult, but there is hope!

Preliminary results from IPCW with four regression methods. Seem to produce fairly similar weights.

Still an argument to be made for adjustment, since some variables were completely separated from the outcome



Results of the unweighted GEE show superiority and non-inferiority, although this is likely attenuated

	t=1	t=2	t=3	t=4	t=5	t=6
Average Treatment Effect at t	1.16	0.64	0.33	0.2	0.01	-0.09
Robust SE (interaction)	0.146	0.246	0.252	0.265	0.264	0.274

At Day 1 PM, patients taking opioids on average report 1.16 points higher on worst-pain on an NRS 0-10 scale than patients taking non-opioids